

KENDRIYA VIDYALAYA SANGATHAN
REGIONAL OFFICE SILCHAR
PRE- BOARD EXAM-1 QUESTION PAPER
CLASS- X (2024-25)
MATHEMATICS STANDARD (041)

SET-1

TIME: 3 hours

MAX.MARKS: 80

General Instructions:

Read the following instructions carefully and follow them:

1. This question paper contains 38 questions.
2. This Question Paper is divided into 5 Sections A, B, C, D and E.
3. In Section A, Questions no. 1-18 are multiple choice questions (MCQs) and questions no. 19 and 20 are Assertion- Reason based questions of 1 mark each.
4. In Section B, Questions no. 21-25 are very short answer (VSA) type questions, carrying 02 marks each.
5. In Section C, Questions no. 26-31 are short answer (SA) type questions, carrying 03 marks each.
6. In Section D, Questions no. 32-35 are long answer (LA) type questions, carrying 05 marks each.
7. In Section E, Questions no. 36-38 are case study based questions carrying 4 marks each with sub parts of the values of 1, 1 and 2 marks each respectively.
8. All Questions are compulsory. However, an internal choice in 2 Questions of section B, 2 Questions of section C and 2 Questions of section D has been provided. And internal choice has been provided in all the 2 marks questions of Section E.
9. Draw neat and clean figures wherever required.
10. Take $\pi = 22/7$ wherever required if not stated.
11. Use of calculators is not allowed.

SECTION-A

This section consists of 20 questions of 1 mark each.

Q.1 The ratio of HCF to LCM of the least composite number and the least prime number is

- (a) 1:2 (b) 2:1 (c) 1:1 (d) 1:3

Q.2 If α and β are zeroes of the polynomial $x^2 - 1$, then value of $(\alpha + \beta)$ is

- (a) 2 (b) 1 (c) -1 (d) 0

Q.3 The quadratic polynomial, the sum of whose zeroes is -5 and their product is 6, is

- (a) $x^2 + 5x + 6$ (b) $x^2 - 5x + 6$ (c) $x^2 - 5x - 6$ (d) $-x^2 + 5x + 6$

Q.4 The zeroes of the polynomial $3x^2 + 11x - 4$ are:

- (a) $\frac{1}{2}, -4$ (b) $\frac{1}{4}, -3$ (c) $\frac{1}{3}, -4$ (d) $\frac{1}{3}, -4$

Q.5 Two lines are given to be parallel. The equation of one of the lines is $3x - 2y = 5$, the equation of the second line can be

- (a) $9x + 8y = 7$ (b) $-12x - 8y = 7$ (c) $-12x + 8y = 7$ (d) $12x + 8y = 7$

Q.6 The values of k for which the equation $2x^2 - kx + 1 = 0$ has real and equal roots is/are:

- (a) $2\sqrt{2}$ (b) $\pm 2\sqrt{2}$ (c) $-2\sqrt{2}$ (d) 2

Q.7 The first term of an AP is p and the common difference is q , then its 10th term is

- (a) $q+9p$ (b) $p+9q$ (c) $p-9q$ (d) $2p+9q$

Q.8 The distance of the point $(-6, 8)$ from origin is :

- (a) 6 (b) -6 (c) 8 (d) 10

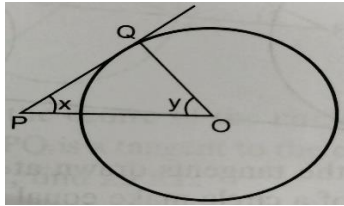
Q.9 The distance between the points $(a \cos \theta + b \sin \theta, 0)$ and $(0, a \sin \theta - b \cos \theta)$ is

- (a) $a^2 + b^2$ (b) $a^2 - b^2$ (c) $\sqrt{a^2 + b^2}$ (d) $\sqrt{a^2 - b^2}$

Q.10 The ratio in which the line $3x+y-9=0$ divides the line segment joining the points $(1,3)$ and $(2,7)$ is

- (a) 3:2 (b) 2:3 (c) 4:3 (d) 3:4

Q.11 In the given figure, PQ is a tangent to the circle with centre O. If $\angle OPQ = x$, $\angle POQ = y$, then $x+y$ is :



- (a) 45° (b) 90° (c) 60° (d) 180°

Q.12 If $\sin \theta = x$ and $\sec \theta = y$ then find the value of $\cot \theta$

- (a) xy (b) $x+y$ (c) $\frac{x}{y}$ (d) $\frac{1}{xy}$

Q.13 The value of $(\operatorname{cosec}^2 \theta - 1) \cdot \tan^2 \theta$ is

- (a) 1 (b) 0 (c) 2 (d) None of these

Q.14 Two cubes of 7 cm each are kept together joining edge to edge to form a solid. The volume of the resulting solid is

- (a) 700 cm^3 (b) 676 cm^3 (c) 686 cm^3 (d) 400 cm^3

Q.15 Two identical solid hemispheres of equal base diameter 10 cm are stuck together along their bases. The total surface area of the combination is

- (a) $400 \pi \text{ cm}^2$ (b) $200 \pi \text{ cm}^2$ (c) $300 \pi \text{ cm}^2$ (d) $100 \pi \text{ cm}^2$

Q.16 If the Mean and the Mode of a distribution are 20 and 14 respectively, then the Median of the distribution is:

- (a) 18 (b) 17 (c) 16 (d) 15

Q.17 In the following distribution:

Height (in cm)	100-115	115-130	130-145	145-160	160-175
Number of Students	15	13	11	10	11

The sum of the lower limit of the modal class and the upper limit of the median class is

- (a) 230 (b) 260 (c) 245 (d) 275

Q.18 When a die is thrown, the probability of getting a composite number is

- (a) $\frac{1}{2}$ (b) $\frac{1}{3}$ (c) $\frac{2}{3}$ (d) 0

Directions: In question number 19 and 20 a statement of Assertion (A) is followed by a statement of Reason(R) Choose the correct option.

- (a) Both, Assertion(A) and Reason(R) are true and Reason(R) is correct explanation of Assertion (A)
 (b) Both, Assertion(A) and Reason(R) are true and Reason(R) is not correct explanation for Assertion (A)
 (c) Assertion(A) is true but Reason(R) is false.
 (d) Assertion(A) is false but Reason(R) is true.

Q.19 Assertion(A):The probability that a leap year has 53 Sundays is $\frac{2}{7}$.

Reason(R): The probability that a non-leap year has 53 Sundays is $\frac{5}{7}$.

Q.20 Assertion(A): Two players Sania and Sonia play a tennis match. The probability of Sania winning the match is 0.79 and that of Sonia winning the match is 0.21.

Reason(R): The sum of probabilities of two complementary events is 1.

SECTION-B

This section consists of 5 questions of 2 marks each.

Q.21 Given that $\text{HCF}(306, 657) = 9$, find $\text{LCM}(306, 657)$.

OR

Check whether 6^n can end with the digit 0 for any natural number n.

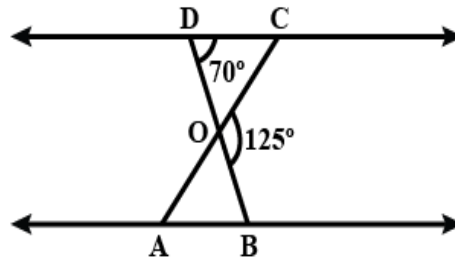
Q.22 Given the linear equation $2x + 3y - 8 = 0$, write another linear equation in two variables such that the geometrical representation of the pair so formed is:

- (i) Intersecting lines
 (ii) Parallel lines

Q.23 E and F are points on the sides PQ and PR respectively of a $\triangle PQR$. State whether $EF \parallel QR$
 $PE = 4$ cm, $QE = 4.5$ cm, $PF = 8$ cm and $RF = 9$ cm

OR

In given figure, $\triangle ODC \sim \triangle OBA$, $\angle BOC = 125^\circ$ and $\angle CDO = 70^\circ$. Find $\angle DOC$, and $\angle OAB$.



Q.24 If $\tan(A + B) = \sqrt{3}$ and $\tan(A - B) = \frac{1}{\sqrt{3}}$; $0^\circ < A + B \leq 90^\circ$; $A > B$, find A and B.

Q.25 Surya Kumar tosses two different coins simultaneously. Find the probability of getting at least one tail.

SECTION-C

This section consists of 6 questions of 3 marks each.

Q.26 Prove that $\sqrt{3}$ is an irrational.

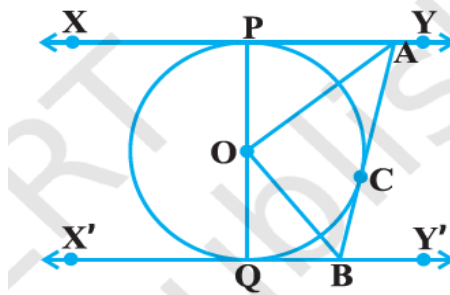
Q.27 A lending library has a fixed charge for the first three days and an additional charge for each day thereafter. Virat paid **Rs.27** for a book kept for **seven days**, while Rohit paid **Rs.21** for the book he kept for **five days**. Find the fixed charge and the charge for each extra day.

Q.28 Find the ratio in which the line segment joining A (1, -5) and B (-4, 5) is divided by the x-axis. Also find the coordinates of the point of division.

OR

Find a point on the y-axis which is equidistant from the points A(6, 5) and B(-4, 3).

Q.29 In figure given below, XY and X'Y' are two parallel tangents to a circle with centre O and another tangent AB with point of contact C intersecting XY at A and X'Y' at B. Prove that $\angle AOB = 90^\circ$.

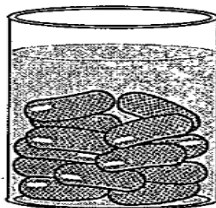


Q.30 Prove that $\frac{\tan A}{1 - \cot A} + \frac{\cot A}{1 - \tan A} = \sec A \cdot \operatorname{cosec} A + 1$

OR

If $\tan \theta + \sin \theta = m$ and $\tan \theta - \sin \theta = n$, prove that $(m^2 - n^2) = 4\sqrt{mn}$

Q.31 A gulab Jamun, contains sugar syrup up to about 30% of its volume. Find approximately how much syrup would be found in 45 gulab jamuns, each shaped like a cylinder with two hemispherical ends with length 5 cm and diameter 2.8 cm.



SECTION-D

This section consists of 4 questions of 5 marks each.

Q.32 Two poles of equal heights are standing opposite each other on either side of the road, which is 80 m wide. From a point between them on the road, the angles of elevation of the top of the poles are 60° and 30° , respectively. Find the height of the poles and the distances of the point from the poles.

Q.33 In a circle of radius 21 cm, an arc subtends an angle of 60° at the centre. Find:

- (i) The length of the arc
- (ii) Area of the sector formed by the arc
- (iii) Area of the segment formed by the corresponding chord.

Q.34 If the 3rd and the 9th terms of an AP are 4 and – 8 respectively, which term of this AP is zero?

OR

If the sum of the first n terms of an AP is $4n - n^2$, what is the first term (that is S_1)? What is the sum of first two terms? What is the second term? Similarly, find the 3rd, the 10th and the n th terms.

Q.35 if a line is drawn parallel to one side of a triangle to intersect the other two sides in distinct points, then prove that the other two sides are divided in the same ratio.

OR

If AD and PM are medians of triangles ABC and PQR, respectively where $\triangle ABC \sim \triangle PQR$, prove that $\frac{AB}{PQ} = \frac{AD}{PM}$

SECTION-E

This section consists of 3 Case –Study Based Questions of 4 marks each.

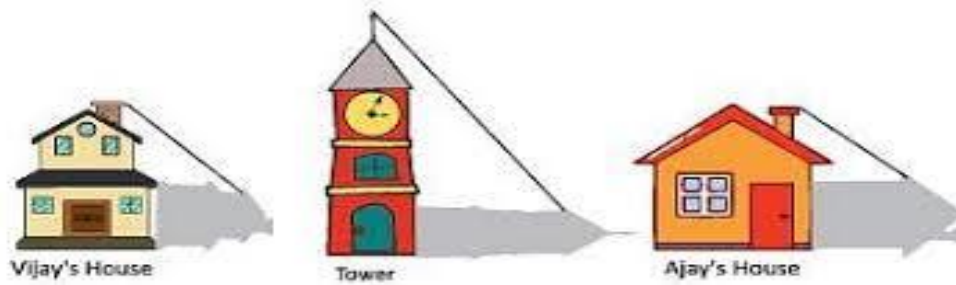
Q.36 Raj and Ajay are very close friends. Both the families decide to go to Shillong by their own cars. Raj's car travels at a speed of x km/h while Ajay's car travels 5 km/h faster than Raj's car. Raj took 4 hours more than Ajay to complete his journey of 400 km.



Based on the above information, answer the following questions:

- (i) What will be the distance covered by Ajay's car in two hours? (in terms of x) (1 mark)
 - (ii) Write a quadratic equation which describes the speed of Raj's car? (1 mark)
 - (iii) What is the speed of Raj's car? (2 marks)
- OR**
- How much time took Ajay to travel 400 km?

Q.37 Vijay is trying to find the average height of a tower near his house. He is using the properties of similar triangles. The height of Vijay's house is 20 m when Vijay's house casts a shadow 10 m long on the ground. At the same time, the tower casts a shadow 50 m long on the ground and the house of Ajay casts 20 m shadow on the ground.



Based on the above information, answer the following questions:

- (i) What is the height of the tower? (1 mark)
- (ii) What will be the length of the shadow of the tower when Vijay's house casts a shadow of 12 m? (1m mark)
- (iii) When the tower casts a shadow of 40 m, same time what will be the length of the shadow of Ajay's house? (2 marks)

OR

When the tower casts a shadow of 40 m, same time what will be the length of the shadow of Vjay's house?

Q.38 During the annual sports meet in a school, all the athletes were very enthusiastic. They all wanted to be the winner so that their house could stand first. The instructor noted down the time taken by a group of students to complete a certain race. The data recorded is given below:

Age(in years)	Number of Students
0-20	1
20-40	4
40-60	3
60-80	7
80-100	5



Based on the above information, answer the following questions:

- (i) What is the class mark of the modal class? (1 mark)
- (ii) Find the median class of the given data. (1 mark)
- (iii) Find the mode of the given data. (2 marks)

OR

Find the median time (in seconds) of the given data.

